

KIERAN RINGEL

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EDUCATION

PhD: University of California Davis - in progress

Davis, CA

Computer Science

September 2025 - Present

Bachelors of Science: Montana State University

Bozeman, MT

Major: Computer Science

August 2018 - May 2022

Minor: Computer Engineering

Honors College Highest Distinction

GPA 3.76/4.0

RESEARCH EXPERIENCE

Computational Climate and Ocean Group

Davis, CA

PhD Student

September 2025 - Current

Los Alamos National Lab

Los Alamos, NM

Center for Nonlinear Studies Post-Baccalaureate Researcher

May 2024 - June 2025

- Learned about parallel computing techniques and HPC systems through the Parallel Computing Summer Research Internship.
- Collaborating on the development of Omega, an ocean model in C++, by learning how to code PDEs and doing performance testing of the model on multiple HPC systems on CPUs and GPUs.
- Developing an emulator of an ice sheet calving model to be included in a the larger DOE ice sheet model using a graph neural networks and convolutional neural networks.
- Summer 2024 funding through DOE Science Undergraduate Laboratory Internships program

Bradley Whitaker Lab

Bozeman, MT

Billings Clinic Covid-19 Forecasting

September 2020 - May 2021

- Partnered with the Billings Clinic to help improve their TensorFlow neural network that predicted Covid cases up to a week out for each county in Montana by extending the architecture to add static variables.

Los Alamos National Lab

Los Alamos, NM

Data Science at Scale Summer School

June 2020 - August 2020

- Created a driving video story of climate models produced by E3SM scientists in collaboration with two computer scientists, five climate scientists and two artists.
- Used ParaView connected to LANL super computers to process hundred gb climate model files and produce state files for visualizations as well as ParaView python scripts to create content for the video.

Hammonds Research Group

Bozeman, MT

Structural Undergraduate Researcher

January 2019 - April 2020

- Performed research on soluble impurities effect on the mechanical properties of ice in the Subzero Research Laboratory.
- Performed compression tests to observe the stress vs strain curve of samples with different levels of sulfuric acid.
- Micro computed x-ray tomography was performed to provide additional information about the micro-structural evolution of samples.

WORK EXPERIENCE

Microsoft

Redmond, WA

Software Engineer - Bing Ads Platform Payment Solutions Team

August 2022 - March 2024

- Collaborated as a backend developer within an agile development team of 3 members. Created an integrated e-commerce solution using C#, by completing end to end integration of external APIs. This demonstrated it's functionality to expand to products beyond the immediate team's scope.
- Contributed to the development of and end to end subscription service in C++.

- Assumed the role of the designated responsible individual (on-call) on a rotational basis. Investigated anomalies on the Grafana dashboard, utilizing the Kusto query language. This ensured the stability and reliability of critical systems.

Yellowstone Ecological Research Center (YERC)

Bozeman, MT

Spatial Data Ingestion Capstone Project

September 2021 - May 2022

- In partnership with another student, we developed an ETL program using the SpatioTemporal Asset Catalog API to get data for YERC's data sharing platform from the Microsoft Planetary Computer.
- Wrote a design document to propose solution to YERC and worked closely with the CTO of YERC to change specifications as the company's needs changed.

Micron Technology

Boise, ID

Test Solutions Engineering Intern

May 2021 - August 2021

- Worked within a team of 10 engineers to increase the efficiency of testing DRAM wafers.
- Simplified a large data set of tests to show test overlaps across thousands of data points in python using Pandas.
- Created a tableau visualization used by other team members to quickly determine which tests overlap and therefore if a test is worth the resources to run.

TEACHING EXPERIENCE

Montana State University

Bozeman, MT

Computer Science Undergraduate Course Assistant CS 145 - Web Design

August 2020 - April 2021

- Independently led a lab of 30 students in the CS and graphic design departments through projects to introduce HTML and CSS concepts.

PUBLICATIONS

- The Ocean Model for E3SM Global Applications: Omega Version 0.1.0. A New High-Performance Computing Code for Exascale Architectures
Petersen, M. R, **Ringel, K.K.**, et al
under review at Geosci. Model Dev.

CONFERENCE PUBLICATIONS

- Scientific Visualization of Antarctica Ice Sheet and Southern Ocean Evolution
Patchett, J.M., Samsel, F., **Ringel, K.**, Abram, G.D., Woods, N., Lipsa, D., Hoffman, M.J., Price, S.F., Petersen, M.R., Zhang, T. and Asay-Davis, X.S.
Super Computing 2020

POSTERS

- GPU/CPU Performance Results on Exascale Architectures for OMEGA: The Ocean Model for E3SM Global Applications
Ringel, K., Asay-Davis, X.S., Begeman, C., Bishnu, S., Brus, S., Jones, P., Kang, H., Kim, Y., Mametjanov, A., O'Neill, B., Petersen, M., Van Roekel, L., Sreepathi, S., Mawuszewski, M.
American Meteorological Society Annual Meeting 2025
- Ecosystem Prognosis Impact Information Cooperative (EPIIC)
Ringel, K., Madsen, H., Izurieta, C., Roberts, R.
MSU Spring Student Research Celebration 2022
- Neural Networks Applied to the COVID-19 Pandemic
Ringel, K., Boyd, I., Hedges, D., Whitaker, B.
National Conference for Undergraduate Research 2021
- Soluble Impurities Effect on Mechanical Abilities of Snow and Ice
Ringel, K., Schrer, E., Hammonds, K.
National Conference for Undergraduate Research 2020 (accepted but did not present after cancellation due to COVID-19)

AWARDS

- 2025 NSF Graduate Research Fellowship Program Honorable Mention
- 2024 Department of Energy Science Undergraduate Laboratory Internship (SULI) Award
- 2nd Place Super Computing 2020 Scientific Visualization & Data Analytics Showcase
- 2018-2022 Montana State University Presidential Scholar - full tuition and \$4,000/yr stipend
- 2018-2022 Montana State University Dean's List - 8 semesters